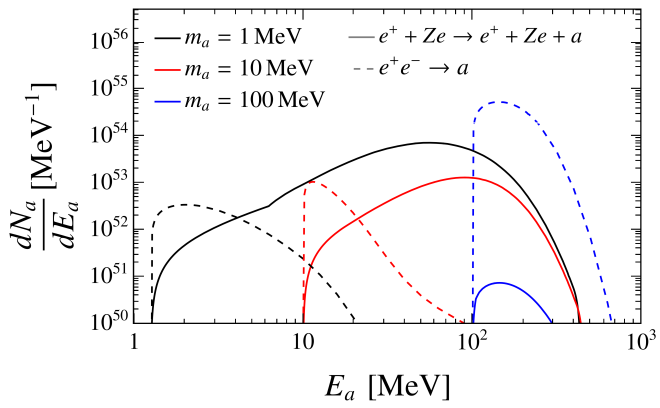

FIG. 1. SNe ALP production spectra via electron coupling g_{ae} integrated over the SN volume and over the duration of the burst. Solid lines depict electron bremsstrahlung emission spectra, while electron fusion spectra are displayed a dashed lines. The different colors refer to different ALP masses. In this figure we set $g_{ae} = 7.5 \times 10^{-10}$.

where where $E_{-,+}$ and $f_{-,+}$ are the electron and positron energies and distribution functions, respectively, while

$$E_{\rm min,max} = \frac{\omega_a}{2} \pm \frac{\sqrt{\omega_a^2 m_a^2 + m_a^4 - 2|\mathbf{p}_a|^2 m_e^2}}{2m_a}\,, \quad (7)$$

in which $\mathbf{p}_a$ is the ALP momentum.

Fig. 1 displays SN emission spectra for ALPs coupled to electrons integrated over the whole SN volume and over the duration of the SN cooling phase $\Delta t \simeq 10$ s. We observe that electron bremsstrahlung is characterized by quasi-thermal spectra peaking at energies $E \simeq m_a + \frac{3}{2}\,T$, where $T \simeq 30\,$MeV is the inner SN core temperature at the beginning of the cooling phase. In particular, the reduction in bremsstrahlung emission for $m_a = 100\,$MeV is related to the strong Boltzmann suppression observed for ALPs with masses $m_a \gtrsim 3T$. Conversely, as discussed in Ref. [57], the electron fusion contribution is suppressed for $m_a \lesssim 20\,$MeV while it becomes dominant only for higher masses. This trend is related to the fact that low-mass ALP never meet the threshold condition for this process $m_a \gtrsim 2\,m_e^*$ for effective electron masses observed in the inner core during the first instants of the cooling phase $m_e^* \sim 10\,$MeV. Therefore, ALP emission through electron fusion plays a role only at $t_{\rm pb} \gtrsim 3$ s, when the SN core is colder and ALP production dramatically decreases. On the other hand, electron fusion production is active for the whole duration of the SN cooling phase for ALPs with masses $m_a \gtrsim 30\,$MeV, resulting in a larger contribution compared to electron bremsstrahlung.

ALPs coupled to nucleons and electrons

Building upon the analysis developed for ALPs coupled to electrons only, we now consider the physics case of ALPs interacting with both nucleons and electrons [28]. This phenomenological scenario is of definite interest for our study, since a UV complete model accounting for ALP-lepton couplings naturally includes also ALP-quark couplings. Then, at energies below the confinement scale, interactions with the quark content of QCD induce ALP couplings to nuclear matter, which are typically described by the following Lagrangian [58, 59]

$$\begin{aligned}\mathcal{L}_{\rm int} = g_a \frac{\partial_\mu a}{2m_N}\Bigg[& C_{ap}\bar{p}\gamma^\mu\gamma_5 p + C_{an}\bar{n}\gamma^\mu\gamma_5 n + \\ & + \frac{C_{a\pi N}}{f_\pi}(i\pi^+\bar{p}\gamma^\mu n - i\pi^-\bar{n}\gamma^\mu p) + \\ & + C_{aN\Delta}\left(\bar{p}\,\Delta^+_\mu + \overline{\Delta^+_\mu}\,p + \bar{n}\,\Delta^0_\mu + \overline{\Delta^0_\mu}\,n\right)\Bigg]\,, \end{aligned} \quad (8)$$

where $g_a = m_N/f_a$ is the dimensionless axion-nucleon coupling, $m_N = 938$ MeV is the nucleon mass, C_{aN} with $N = p, n$ are model-dependent $\mathcal{O}(1)$ coupling constants, $f_\pi = 92.4$ MeV is the pion decay constant, $C_{a\pi N} = (C_{ap} - C_{an})/\sqrt{2}g_A$ [60] is the axion-pion-nucleon coupling and $C_{aN\Delta} = -\sqrt{3}/2\,(C_{ap} - C_{an})$ is the axion-nucleon-Δ baryon coupling, with $g_A \simeq 1.28$ [61] the axial coupling. For convenience, we define the axion-proton and axion-neutron coupling as $g_{aN} = g_a C_{aN}$ for $N = p, n$. Inspired by the Kim-Shifman-Vainshtein-Zakharov (KSVZ) axion model [62], here we set $g_{an} = 0$.

If ALPs are coupled to nuclear matter, then nuclear interactions are the main channel for ALP production in the hot and dense SN core, which is almost entirely constituted by neutrons and protons. In particular, the interaction Lagrangian in Eq. (8) allows for two competing processes: nucleon-nucleon (NN) bremsstrahlung $NN \to NNa$, and pionic Compton-like scatterings (also called pion conversions) $\pi N \to Na$. The computation of the axion emissivity via NN bremsstrahlung has been investigated in a number of works on the topic [1, 63–65] and the state-of-the-art calculation for the related axion emission rate has been introduced in Ref. [8] accounting for a non-vanishing mass for the pion exchanged by the interacting nucleons [66], the contribution from the two-pion exchange [67], effective in-medium nucleon masses and multiple nucleon scattering effects [68, 69]. Interestingly, Ref. [70] has demonstrated that finite density effects on axion-nucleon interactions could magnify the bremsstrahlung emission rate. On the other hand, the impact of pion conversion processes [6, 65, 71] has been recently reevaluated in Ref. [9], since the authors of Ref. [72] pointed out that strong interactions can magnify the abundance of negatively-charged pions. Under this assumption, the pion conversion emission rate could become comparable and even dominant over the bremsstrahlung contribution. Furthermore, Refs. [60, 73] have recently shown that the contribution from the contact interaction term and the $\Delta(1232)$ resonance significantly enhance